Abstract

We present an open-source Python implementation for polyhedral full-tensor gravity forward modelling based on a numerically stable analytical formulation. The code leverages Numba just-in-time compilation to achieve high performance while retaining the simplicity and flexibility of Python. Two algorithmic variants are provided: one precomputes face and edge geometry for optimal speed with moderate mesh sizes, and another computes geometric quantities on-the-fly to minimize memory usage for very large models. The implementation supports computation of the gravitational potential, vector, and full gradient tensor, and is validated through multiple benchmark tests—including idealized spheres, asteroid shape models, and lunar topography—demonstrating excellent accuracy, robustness, and convergence across diverse scenarios.

Introduction

Polyhedral models have become a cornerstone in gravitational and magnetic field forward modelling due to their unique ability to represent sources with arbitrarily complex geometries-ranging from local topographic features to planetary-scale bodies such as asteroids and moons. Unlike simpler discretization schemes like rectangular prisms or tesseroids, which often introduce artificial discontinuities at interfaces, polyhedral models preserve the continuity and roughness of natural density or magnetization boundaries, thereby offering superior accuracy-particularly in the near-zone where high-frequency gravitational signals dominate (Tsoulis, 2001; Hikida and Wieczorek, 2007; Benedek et al., 2018; Zhang and Chen, 2018; Hirt et al., 2019; Holzrichter et al., 2019; Saraswati et al., 2019).

Over the past two decades, significant progress has been made in deriving space-domain analytical solutions for the gravitational potential, vector, and gradient tensor generated by polyhedral bodies. Early foundational work established closed-form expressions for homogeneous or linearly varying density distributions (Barnett, 1976; Okabe, 1979; Holstein and Ketteridge, 1996; Werner and Scheeres, 1996; Pohánka, 1998; Tsoulis and Petrović, 2001; Holstein, 2003; Tsoulis, 2012), More recently, these formulations have been generalized to accommodate arbitrary polynomial density functions, enabling more realistic modelling of geophysical sources with heterogeneous internal structures (D'Urso and Trotta, 2017). In particular, a series of works by Ren et al. (2017, 2018, 2020, 2022) unified decades of prior results into a comprehensive analytical framework for both gravitational and magnetic fields of polyhedra with polynomial density or magnetization variations.

Beyond space-domain approaches, spectral methods have also played an important role. Jamet and Tsoulis (2020) introduced a novel and numerically stable method for computing spherical harmonic coefficients of the gravity field of a constant-density polyhedron by reducing volume integrals of solid harmonics to line integrals along the polyhedron's edges. Spherical harmonic transform (SHT) based techniques, initiated by Wieczorek and Phillips (1998) as a spherical analog to the Cartesian method of Parker (1973), enable efficient global topographic gravity modelling. However, solid spherical harmonic series suffer from divergence inside the Brillouin sphere-even for near-spherical large planets like Earth and Moon at high degrees-and are ill-suited for highly irregular bodies such as asteroids and comets (Hu and Jekeli, 2015; Reimond and Baur, 2016; Hirt and Kuhn, 2017; Hirt et al., 2019; Bucha et al., 2019; Šprlák and Han, 2021); spheroidal/ellipsoidal harmonics offer improved convergence at the expense of more complex parametrization and higher computational cost (Wang and Yang, 2013; Claessens and Hirt, 2013; Rexer et al., 2016; Liu et al., 2025).

Fourier-domain methods based on Cartesian Fast Fourier Transform (FFT) algorithms have also received significant attention in solving local modelling problems when a flat-earth approximation is applicable. Analytical wavenumber-domain formulations by Pedersen (1978), Wu (1983), and Hansen and Wang (1988) laid the foundation for FFT-based polyhedral gravity forward modelling. Later, Wu (2019); Wu et al. (2019) developed efficient 2D/3D Fourier-domain algorithms that recast polyhedral gravity forward modelling-for bodies with constant or exponential density-into wavenumber-domain line integrals, leveraging Gauss-FFT (Wu and Tian, 2014) and nonuniform FFT (NUFFT) (Greengard and Lee, 2004; Keiner et al., 2009) techniques to achieve high computational accuracy, though computational cost still grows rapidly with model resolution due to the analytical evaluation of the spectrum. Recent adaptations of Parker's method to general polyhedra further demonstrate the potential of spectral efficiency Wu (2021), yet space-domain formulations remain essential for validation and high-precision applications.

Accompanying these theoretical advances, several open-source implementations have become available. These include a FORTRAN implementation by Tsoulis (2012), MATLAB routines introduced by Singh and Guptasarma (2001) and later extended by Saraswati et al. (2019), and a C++ library (`https://github.com/zhong-yy/GraPly`) that implements the algorithms of Ren et al. (2017, 2018, 2020, 2022), and a recently released C++ library with a Python interface based on the line integral formulation of Tsoulis (2012) (Schuhmacher et al., 2024). However, practical use of existing implementations has revealed persistent numerical instabilities—not only when evaluation points lie on or near polyhedral faces or vertices, but occasionally even at seemingly benign locations (these include application of Tsoulis (2012) FORTRAN code in Wu (2016), Ren et al. (2018) C++ code in Zhou et al. (2020), and Saraswati et al. (2019) MATLAB code in modelling Bennu asteroid, personal communication). These observations underscore the ongoing need for a robust implementation that guarantees consistent accuracy and reliability across all configurations.

In this paper, we provide an open-source, pure-Python implementation for polyhedral gravity forward modelling based on the method introduced by Werner and Scheeres (1996), which we find to be numerically stable, concise, and elegant. Critically, our implementation leverages Numba just-in-time (JIT) compilation to achieve performance comparable to compiled languages while maintaining Python's accessibility and ease of use. Two algorithmic variants are developed: one constructs an edge list from the face–vertex geometry and precomputes face and edge normals in memory to accelerate computations for meshes up to $N_T \lesssim 10^8$ triangular facets; the other computes all geometric quantities on-the-fly, requiring significantly less memory and thus better suited for very large polyhedral meshes with $N_T \gg 10^8$ facets. The Numba acceleration enables efficient parallel processing across observation points and geometric elements, making our code suitable for both local-scale problems and planetary-scale applications with millions to hundreds of millions of mesh elements. Our implementation aims to serve both as a practical tool for geophysical and planetary applications and as a reliable reference for validating emerging spectral or data-driven (Martin and Schaub, 2022) methods.

Methodology

Theoretical Framework

We adopt the analytical formulation for the gravitational potential (GP), gravitational vector (GV), and gravity gradient tensor (GGT) of a homogeneous polyhedron as derived by Werner and Scheeres (1996). This approach expresses all field components in terms of contributions from faces and edges, enabling efficient and numerically stable computation.

We assume the polyhedron is represented as a closed, manifold triangular mesh, composed of a set of vertices $\mathcal{V}$, a set of triangular faces $\mathcal{T}$, and a set of edges $\mathcal{E}$. Let $N_V = |\mathcal{V}|$, $N_T = |\mathcal{T}|$, and $N_E = |\mathcal{E}|$ denote their cardinalities. For a simply connected polyhedron, Euler's polyhedral formula gives $N_V + N_T - N_E = 2$, and since each triangle has three edges and each interior edge is shared by two faces, we have $N_E = 1.5N_T$ and $N_V \approx 0.5N_T$, so all geometric complexities scale linearly with N_T.

For a polyhedron of constant density ρ, the gravitational potential V, gravitational vector $\mathbf{g} = \nabla V$, and gravity gradient tensor $\mathbf{T} = \nabla\nabla V$ at an observation point P are given by Werner and Scheeres (1996):

$$V = \frac{1}{2}G\rho \sum_{e\in\mathcal{E}} \mathbf{r}_e \cdot \mathbf{E}_e \cdot \mathbf{r}_e \, L_e - \frac{1}{2}G\rho \sum_{t\in\mathcal{T}} \mathbf{r}_t \cdot \mathbf{F}_t \cdot \mathbf{r}_t \, \omega_t, \tag{1}$$

$$\mathbf{g} = \nabla V = -G\rho \sum_{e\in\mathcal{E}} \mathbf{E}_e \cdot \mathbf{r}_e \, L_e + G\rho \sum_{t\in\mathcal{T}} \mathbf{F}_t \cdot \mathbf{r}_t \, \omega_t, \tag{2}$$

$$\mathbf{T} = \nabla\nabla V = G\rho \sum_{e\in\mathcal{E}} \mathbf{E}_e \, L_e - G\rho \sum_{t\in\mathcal{T}} \mathbf{F}_t \, \omega_t, \tag{3}$$

where $G = 6.6743 \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$ is the gravitational constant. The vectors $\mathbf{r}_e$ and $\mathbf{r}_t$ denote position vectors from the observation point P to representative points on edge e and triangular face t, respectively. The geometric weighting factors L_e and ω_t encode the influence of each edge and face, while $\mathbf{E}_e$ and $\mathbf{F}_t$ are dyadic tensors constructed from face and edge normals.

The face dyad $\mathbf{F}_t$ for a triangular face $t \in \mathcal{T}$ (oriented counter-clockwise with respect to the outward normal) is defined as:

$$\mathbf{F}_t = \hat{\mathbf{n}}_t\,\hat{\mathbf{n}}_t, \tag{4}$$

where $\hat{\mathbf{n}}_t$ is the unit outward-pointing normal vector of face t.

Each interior edge $e \in \mathcal{E}$ is shared by two adjacent triangular faces t^+ and t^-, denote unit vectors along edge e as $\hat{\mathbf{e}}^+$ and $\hat{\mathbf{e}}^-$ ($\hat{\mathbf{e}}^+ = -\hat{\mathbf{e}}^-$), which are CCW in t^+ and t^-, respectively. The corresponding edge dyad $\mathbf{E}_e$ is:

$$\mathbf{E}_e = \hat{\mathbf{n}}_t^+\,\hat{\mathbf{n}}_e^+ + \hat{\mathbf{n}}_t^-\,\hat{\mathbf{n}}_e^-, \tag{5}$$

where $\hat{\mathbf{n}}_e^+$ and $\hat{\mathbf{n}}_e^-$ are unit edge-normal vectors lying in the planes of their respective faces and perpendicular to the edge direction:

$$\hat{\mathbf{n}}_e^+ = \hat{\mathbf{e}}^+ \times \hat{\mathbf{n}}_t^+, \quad \hat{\mathbf{n}}_e^- = \hat{\mathbf{e}}^- \times \hat{\mathbf{n}}_t^-.$$

The scalar geometric factors depend only on the relative geometry between the observation point P and the polyhedral element. For an edge AB, the per-edge factor L_e is:

$$L_e = \ln\left(\frac{r_{PA} + r_{PB} + e_{AB}}{r_{PA} + r_{PB} - e_{AB}}\right), \tag{6}$$

where $r_{PA} = \|\mathbf{r}_{PA}\|$, $r_{PB} = \|\mathbf{r}_{PB}\|$, and $e_{AB} = \|\mathbf{r}_{AB}\|$ are the Euclidean distances from P to vertices A and B, and the length of edge AB, respectively.

For a triangular facet ABC, the per-face solid angle factor ω_t is:

$$\omega_t = 2\arctan\left(\frac{\mathbf{r}_{PA} \cdot (\mathbf{r}_{PB} \times \mathbf{r}_{PC})}{r_{PA}r_{PB}r_{PC} + r_{PA}(\mathbf{r}_{PB} \cdot \mathbf{r}_{PC}) + r_{PB}(\mathbf{r}_{PC} \cdot \mathbf{r}_{PA}) + r_{PC}(\mathbf{r}_{PA} \cdot \mathbf{r}_{PB})}\right), \tag{7}$$

which represents the signed solid angle subtended by the face at point P.

This formulation provides a unified, closed-form description of the full gravity tensor (potential, vector, and gradient) and serves as the foundation for our efficient and robust Python implementation.

Algorithm Design and Benchmark

We provide two Numba-accelerated implementations of the Werner and Scheeres (1996) polyhedral gravity formulation, tailored for triangular meshes $(\mathcal{V}, \mathcal{T})$. Both compute the gravitational potential V, vector $\mathbf{g}$, and gradient tensor $\mathbf{T}$ from vertex coordinates and face indices, with input coordinates in km, density ρ in kg/m^3, and outputs scaled to geophysical units (V in m^2/s^2, $\mathbf{g}$ in mGal, and $\mathbf{T}$ in Eotvos).

Version 1: Precomputed Geometry

This variant precomputes all static geometric data once per mesh:

- Constructs a global edge list $\mathcal{E}$ storing adjacent faces and orientation.
- Computes unit face normals $\hat{\mathbf{n}}_t$ for all $t \in \mathcal{T}$ and edge normals $\hat{\mathbf{n}}_e^{\pm}$ for all $e \in \mathcal{E}$ in parallel.

During evaluation, each observation point accumulates contributions from solid angles ω_t (faces) and logarithmic edge factors L_e (edges). This minimizes arithmetic overhead and is optimal for repeated evaluations or moderate meshes ($N_T \lesssim 10^8$), at the cost of approximately $164\,N_T$ bytes of memory (e.g., $\sim$16 GB for $N_T = 10^8$).

Algorithm 1 WerSch_numba_v1 (Precomputed Geometry)

Require: Observation points $\mathbf{P}$, vertices $\mathcal{V}$, faces $\mathcal{T}$, density ρ
Ensure: $V(\mathbf{P})$, $\mathbf{g}(\mathbf{P})$, $\mathbf{T}(\mathbf{P})$ in geophysical units

$\mathcal{E} \leftarrow$ BuildEdgeList($\mathcal{T}$)
$(\hat{\mathbf{n}}_t)_{t\in\mathcal{T}} \leftarrow$ ComputeFaceNormals($\mathcal{V}, \mathcal{T}$)
$(\hat{\mathbf{n}}_e^+, \hat{\mathbf{n}}_e^-)_{e\in\mathcal{E}} \leftarrow$ ComputeEdgeNormals($\mathcal{V}, \mathcal{E}, \hat{\mathbf{n}}_t$)
for all $\mathbf{p} \in \mathbf{P}$ **do in parallel via Numba** `prange`
    Initialize accumulators for V, $\mathbf{g}$, $\mathbf{T}$
    for all $t \in \mathcal{T}$ **do**
        Compute solid angle $\omega_t(\mathbf{p})$
        Accumulate face contribution to V, $\mathbf{g}$, $\mathbf{T}$
    end for
    for all $e \in \mathcal{E}$ **do**
        Compute logarithmic factor $L_e(\mathbf{p})$
        Accumulate edge contribution to V, $\mathbf{g}$, $\mathbf{T}$
    end for
    Apply unit scaling to geophysical conventions
end for

Version 2: On-the-Fly Geometry

To reduce memory footprint, all geometric quantities are computed during evaluation:

- Face normals $\hat{\mathbf{n}}_t$ are computed per triangle.
- Edge normals are derived on the fly via $\hat{\mathbf{n}}_e = \hat{\mathbf{e}} \times \hat{\mathbf{n}}_t$.

No global edge list is stored; each interior edge is processed twice (once per adjacent face), with contributions accumulating correctly due to the linearity of the formulation. Memory usage is only $\sim 24\,N_T$ bytes (e.g., ~ 2.4 GB for $N_T = 10^8$), and can be further reduced by partitioning $\mathcal{T}$ into chunks-ideal for extreme-scale models ($N_T \gg 10^8$).

Algorithm 2 WerSch_numba_v2 (On-the-Fly Geometry)

```
Require: P, 𝒱, 𝒯, ρ
Ensure: V(P), g(P), T(P) in geophysical units
for all p ∈ P do in parallel via Numba prange
    Initialize accumulators for V, g, T
    for all t ∈ 𝒯 do
        Compute n̂_t
        Compute ω_t(p), accumulate face terms
        for all edge e of triangle t do
            Compute ê
            Compute n̂_e ← ê × n̂_t
            Compute L_e(p), accumulate edge terms
        end for
    end for
    Apply unit scaling to geophysical conventions
end for
```

Both versions employ numerically stable evaluation (`atan2`-based solid angle computation, safeguarded logarithms for L_e) and leverage Numba's parallel JIT compilation. Version 1 is recommended for speed-critical scenarios with repeated evaluations, and Version 2 is more suitable for memory-constrained or large-scale modelling where $N_T \gg 10^8$.

The performance of both algorithmic variants critically depends on Numba's just-in-time (JIT) compilation, which transforms Python loops into optimized native machine code with minimal overhead. Without this acceleration, the nested loops over millions of faces, edges, and observation points would be prohibitively slow in pure Python. Numba enables efficient vectorization, cache-friendly memory access, and automatic parallelization across CPU cores via the `prange` construct—making our implementations competitive with compiled-language alternatives while retaining Python's readability and rapid prototyping advantages. This acceleration is especially vital for Version 2, where on-the-fly geometric computations would otherwise dominate runtime; Numba ensures these dynamic calculations remain efficient even at extreme scales.

We benchmark `WerSch_numba_v1` and `WerSch_numba_v2` to validate their numerical equivalence and compare performance. The test uses a homogeneous sphere ($R = 1.0$ km, $\rho = 2670$ kg/m^3) discretized by recursive icosahedral subdivision (Zhang and Chen, 2018; Wu, 2021), with field evaluations at quasi-uniform points on a concentric sphere ($r_{\mathrm{obs}} = 1.1$ km). This creates a matrix of problems from $N_T, N_P = 20$ up to $\approx 1.3 \times 10^6$.

As shown in Figure 1, the results confirm that both algorithms are numerically identical for practical purposes. The maximum absolute differences are negligible: for the vertical gravity component (g_z), they range from 10^{-14} to 10^{-11} mGal against a reference standard deviation of 35.62 mGal; for the vertical gradient T_{zz}, differences are between 10^{-13} and 10^{-10} Eotvos against a standard deviation of 501.61 Eotvos.

In terms of performance, `v1` is consistently faster for moderate-to-large problems, with the runtime ratio t_2/t_1 stabilizing near 1.3–1.5 for the largest configurations. However, `v2` has a significantly smaller memory footprint due to its on-the-fly computation. This makes `v2` a robust alternative not only for small-scale tests but also for extremely large problems where memory constraints would otherwise prevent the use of precomputed edge-based data structures. Thus, the choice between versions hinges on the trade-off between computational speed (`v1`) and memory efficiency (`v2`).

Numerical examples

An ideal sphere test

First we validate our polyhedral forward modelling implementation using a homogeneous sphere-a canonical test case with a known analytical solution for the gravitational potential, vector, and gradient tensor. We compare two standard approaches for discretizing a spherical surface into triangular polyhedra: recursive icosahedral subdivision (Zhang and Chen, 2018) and triangulation of a regular longitude–latitude (geographic) grid. Both are widely used in geophysical modelling, though they differ in geometric structure, convergence behaviour, and compatibility with existing datasets.

The icosahedral method generates meshes with nearly equilateral triangles and minimal variation in edge lengths across the sphere. As shown in Figure 2 (top row) and quantified in Table 1, this leads to rapid and balanced convergence in both surface area and volume errors as the refinement level n increases. For instance, at $n = 6$ ($N_T = 81920$), the relative area error is already below 10^{-4}, and the ratio of maximum to minimum edge length remains close to 1.5.

In contrast, the geographic grid-while intuitive and aligned with many raster-based geospatial datasets (e.g., digital elevation models, or gravity models on regular grids)-exhibits strong meridian convergence at the poles. This results in highly elongated triangles near the poles and a wide spread in edge lengths, clearly visible in the bottom row of Figure 2. At the same refinement level ($n = 6$), the shortest edges are about two orders of magnitude smaller than the longest ones (see Table 1), which may pose challenges for numerical stability in forward modelling if not properly addressed.

Nevertheless, both discretizations converge systematically toward the ideal sphere. As n increases beyond 8, geometric errors in both area and volume fall below 10^{-5} for both methods (Table 1), confirming that either approach can achieve high geometric fidelity given sufficient resolution. The choice between them thus involves practical trade-offs: icosahedral meshes offer superior geometric regularity and are well-suited for global, high-precision modelling, whereas geographic grids provide straightforward compatibility with structured observational data and simpler indexing for regional applications.

To assess the impact of these geometric differences on gravity forward modelling, we compute the full gravity tensor-comprising the potential V, the gravitational acceleration vector in the North-East-Downward (NED) (local north-oriented coordinate system), and all independent components of the gravity gradient tensor-using our `WerSch_numba_v2` implementation on both mesh types. These results are compared against the analytical solution for a homogeneous sphere of radius $R = 6371.393$ km and density $\rho = 5520$ kg/m^3. Observations are placed at four altitudes above the surface ($h = 0.001$, 0.1, 1, and 10 km), with 5000 quasi-uniform points per altitude generated via Fibonacci sampling.

Figure 3 shows the relative L_2 errors for selected field components as a function of mesh resolution. Both discretizations converge toward the analytical solution as N_T increases, with errors decreasing monotonically across all altitudes. At comparable resolutions, the icosahedral discretization consistently yields marginally lower errors, reflecting its more uniform geometry. As expected, errors are largest in the near-field ($h = 0.001$ km) and diminish rapidly with altitude due to the smoothing effect of the Newtonian potential. Importantly, the gravity field discrepancies align with the underlying geometric approximation errors reported in Table 1: since both the analytical sphere and the polyhedral formulation are exact for their respective domains, the observed differences arise primarily from how closely the polyhedral surface approximates the ideal sphere. This confirms that our implementation operates at machine precision, and that the dominant source of error is geometric-not algorithmic.

EROS shape model test

Next we validate our polyhedral implementation by numerically testing Green's third identity (Blakely, 1995) on the asteroid 433 EROS shape model (Gaskell, 2020). The gravitational potential at any exterior point $\mathbf{r}$ can be reconstructed from the potential and its normal derivative on the body's surface S:

$$\begin{aligned} V(\mathbf{r}) &= \frac{1}{4\pi} \oint_S \left[V(\tilde{\mathbf{r}}) \frac{\partial}{\partial \tilde{n}} \left(\frac{1}{|\mathbf{r} - \tilde{\mathbf{r}}|} \right) - \frac{1}{|\mathbf{r} - \tilde{\mathbf{r}}|} \frac{\partial V(\tilde{\mathbf{r}})}{\partial \tilde{n}} \right] dS \\ &\approx \sum_{i=1}^{N_T} s_i \Big[V(\tilde{\mathbf{r}}_i) \mathcal{G}_{\tilde{n}}(\mathbf{r}, \tilde{\mathbf{r}}_i) - \mathcal{G}(\mathbf{r}, \tilde{\mathbf{r}}_i) g_{\tilde{n}}(\tilde{\mathbf{r}}_i) \Big], \end{aligned} \tag{8}$$

where $\mathcal{G}(\mathbf{r}, \tilde{\mathbf{r}}) = \frac{1}{4\pi|\mathbf{r}-\tilde{\mathbf{r}}|}$ is the Green's function, $\mathcal{G}_{\tilde{n}}(\mathbf{r}, \tilde{\mathbf{r}}_i)$ denotes its first-order outer normal derivative evaluated at the center $\tilde{\mathbf{r}}_i$ of triangular facet S_i, s_i is the area of S_i, and $g_{\tilde{n}}(\tilde{\mathbf{r}}_i) = \partial V/\partial \tilde{n}$ is the outward normal component of the gravity vector at $\tilde{\mathbf{r}}_i$. This representation provides a powerful consistency check: if both V and $g_{\tilde{n}}$ are computed accurately on the surface, the reconstructed potential $V(\mathbf{r})$ should converge to the true value as the mesh is refined.

Using the EROS shape model (856 vertices, 1708 faces) and a constant density of $\rho = 2670$ kg/m^3, we compute V and $g_{\tilde{n}}$ at all facet centers via `WerSch_numba_v1`. We then evaluate Equation (8) on three concentric spheres of radii $r = 17.8$, 18.0, and 20.0 km, comparing the result against a reference solution obtained by direct polyhedral forward modelling. The EROS mesh is progressively refined using linear subdivision; at subdivision level n, the number of triangular faces grows as $N_T = 1708 \times 4^n$, reaching $N_T \approx 7 \times 10^6$ at $n = 6$. This refinement process, illustrated in Fig. 4, preserves geometric fidelity while increasing surface resolution.

As expected, the approximation error decreases systematically with refinement, confirming both the correctness of our field computations and the validity of Green's third identity applied to potential fields caused by complex polyhedral bodies. Figure 5 shows the spatial distribution of relative errors on the innermost evaluation sphere ($r = 17.8$ km) for four refinement levels. At coarse resolutions, errors are elevated near the positive and negative x-axis—regions where EROS is most elongated and closest to the evaluation surface. Importantly, this spatial pattern persists even at the highest refinement level: although the magnitude of the relative error drops below 10^{-5}, the residual errors remain concentrated near the tips along $\pm x$. This indicates that the error structure is not an artifact of poor discretization but reflects the intrinsic sensitivity of the near-field potential to local geometry. The global convergence trend is quantified in Fig. 6, where both RMS and maximum relative errors decrease by over three orders of magnitude as the mesh is refined.

Lunar topographic gravitational potential test

Finally, to further validate our polyhedral implementation at planetary scale, we compute the topographic gravitational potential (TGP) of the Moon using both spherical harmonics (SH) and polyhedral (PH) methods. The lunar shape is modelled using the LOLA spherical harmonic model `Moon_shape_719.sh` downloaded from Wieczorek (2024), truncated to degree 359, which corresponds to a Driscoll-Healy (DH2) equi-sampled latitude/longitude grid of resolution $15'$ (arc-minute) (Driscoll and Healy, 1994; Wieczorek and Meschede, 2018). The topographic model is then obtained by subtracting from the shape model a reference radius of $r_{\mathrm{ref}} = 1737.151$ km—the mean value of the degree-359 SH shape model. A constant crustal density of

$\rho = 2560$ kg/m^3 is assumed.

In the SH approach, implemented via `pyshtools` (Wieczorek and Meschede, 2018), the topographic potential is computed up to $n_{\max} = 7$ in the binomial expansion of the topography (Wieczorek and Phillips, 1998), yielding a gravity field expanded to degree $l_{\max} = 2519$. The resulting gravity vector components (g_x, g_y, g_z) are then evaluated on a $15' \times 15'$ grid on the sphere of radius $r_{\text{calc}} = 1748.0$ km, which lies safely above the maximum topography of the applied lunar shape model. Extending $n_{\max}$ to higher orders would further improve the accuracy of the SH method, but is limited by numerical instabilities in `pyshtools` for spherical harmonic expansions beyond degree 2800. We examine results obtained using $n_{\max}$ from 1 to 7; the incremental contribution beyond $n_{\max} = 7$ is negligible, with maximum discrepancies below 1 mGal and standard deviations under 0.1 mGal. Thus, $n_{\max} = 7$ provides a stable and sufficiently accurate SH reference for comparison with our polyhedral solutions.

For the PH method, the same shape model is converted into triangular meshes based on geographic grids with resolutions of $15'$ ($N_T = 2\,070\,720$) and $6'$ ($N_T = 12\,952\,800$). The forward modelling is performed using our `WerSch_numba_v1` algorithm, with observation points placed on the same $15' \times 15'$ grid at $r_{\text{calc}} = 1748.0$ km. The topographic gravity signal is isolated by subtracting the gravity field of a homogeneous sphere of radius r_{ref} and density ρ from the polyhedral solution of the full shape model.

Table 2 summarizes the statistical comparison between the two methods. At 15-arcmin resolution, the PH solution agrees with the SH reference to within 0.7 mGal (north, g_x), 0.8 mGal (east, g_y), and 1.1 mGal (downward, g_z) in standard deviation, with maximum differences of -13.2, -14.8, and -21.5 mGal, respectively. At 6-arcmin resolution, agreement improves markedly across all components: standard deviations drop to $0.1 - 0.2$ mGal and maximum errors fall below 3.4 mGal, confirming that residual discrepancies stem from geometric discretization rather than algorithmic inaccuracies.

Figure 7 visualizes the global distribution of these differences. The top-left panel shows lunar topography, with white triangles indicating locations of the largest g_z discrepancies in the 15-arcmin PH-SH comparison; these sites were selected to be at least 1000 km apart to ensure spatial independence. The top-right panel displays the SH-derived g_z field, while the bottom-left and bottom-right panels show the difference $g_z^{\text{ph}} - g_z^{\text{sh}}$ for the 15- and 6-arcmin PH models, respectively. Since the input shape model is a continuous spherical harmonic representation, the polyhedral method requires a discrete mesh approximation—introducing resolution-dependent errors that diminish as the mesh is refined. The resulting error pattern is spatially coherent and decreases significantly with finer resolution.

To quantify convergence, we evaluate the gravity field at the ten sites labelled A-J in Figure 7, which exhibit the largest $|g_z|$ discrepancies in the 15-arcmin PH-SH comparison. Figure 8 shows the absolute errors in all three gravity components (g_x, g_y, g_z) as a function of polyhedral mesh resolution, ranging from 15 to 1 arcminute. Since the input lunar shape is a continuous spherical harmonic model, its discretization into a polyhedral mesh introduces resolution-dependent geometric errors; as the mesh is refined, these errors generally diminish. The gravity errors decrease monotonically with increasing resolution and fall below 1 mGal (dashed line) for resolutions finer than ~ 5 arcminutes in most cases. However, at the finest resolution (1 arcminute), the error reduction plateaus or slightly increases for a few locations—most notably point A—suggesting that the SH reference solution, truncated at degree $n_{\max} = 7$, may be approaching its intrinsic accuracy limit, which is on the order of 1 mGal at certain sites. These computations employ algorithm `WerSch_numba_v2`, which is optimized for large-mesh, few-observation scenarios. Overall, this behaviour confirms the numerical consistency of our polyhedral implementation and its convergence toward the SH solution as the surface approximation improves.

Conclusion

This work introduces a reliable and efficient Python-based tool for polyhedral gravity forward modelling that addresses long-standing numerical stability issues observed in several existing implementations. Built upon a well-established analytical framework, our code accurately computes the full gravity tensor—potential, vector, and gradient—for arbitrarily complex polyhedral bodies with constant density. The use of Numba acceleration ensures near-compiled performance, making the tool suitable for both small-scale studies and large planetary applications involving hundreds of millions of triangular facets.

Two distinct algorithmic strategies are implemented to accommodate different computational constraints. The first variant, which precomputes and stores all necessary geometric data (face normals, edge normals, and adjacency information), delivers maximum computational efficiency and is ideal for repeated evaluations or moderately sized meshes. The second variant avoids storing edge-level data and instead recomputes geometric quantities during evaluation, drastically reducing memory consumption at the cost of a modest increase in runtime. This makes it particularly well-suited for extreme-scale problems where memory becomes the limiting factor.

Through rigorous validation—including comparisons with analytical solutions for a homogeneous sphere, consistency checks using Green's third identity on the EROS asteroid shape model, and cross-verification against spherical harmonic methods for lunar topographic gravity—we demonstrate that the dominant source of error in our results stems from geometric discretization rather than numerical inaccuracies in the field computation itself. Errors decrease systematically with mesh refinement, confirming both the correctness of the underlying mathematics and the robustness of the implementation.

The software is fully open-source and designed for ease of use, integration, and extension. It provides a trustworthy reference solution for validating new forward modelling techniques—whether spectral, data-driven, or hybrid—and supports a wide range of applications in geophysics and planetary science. Future extensions could include support for variable density distributions and magnetic forward modelling, building naturally on the current architecture.

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Data and Materials Availability

Python code and Jupyter notebooks to reproduce all the numerical results in this work can be found at `https://doi.org/10.6084/m9.figshare.32403024`.

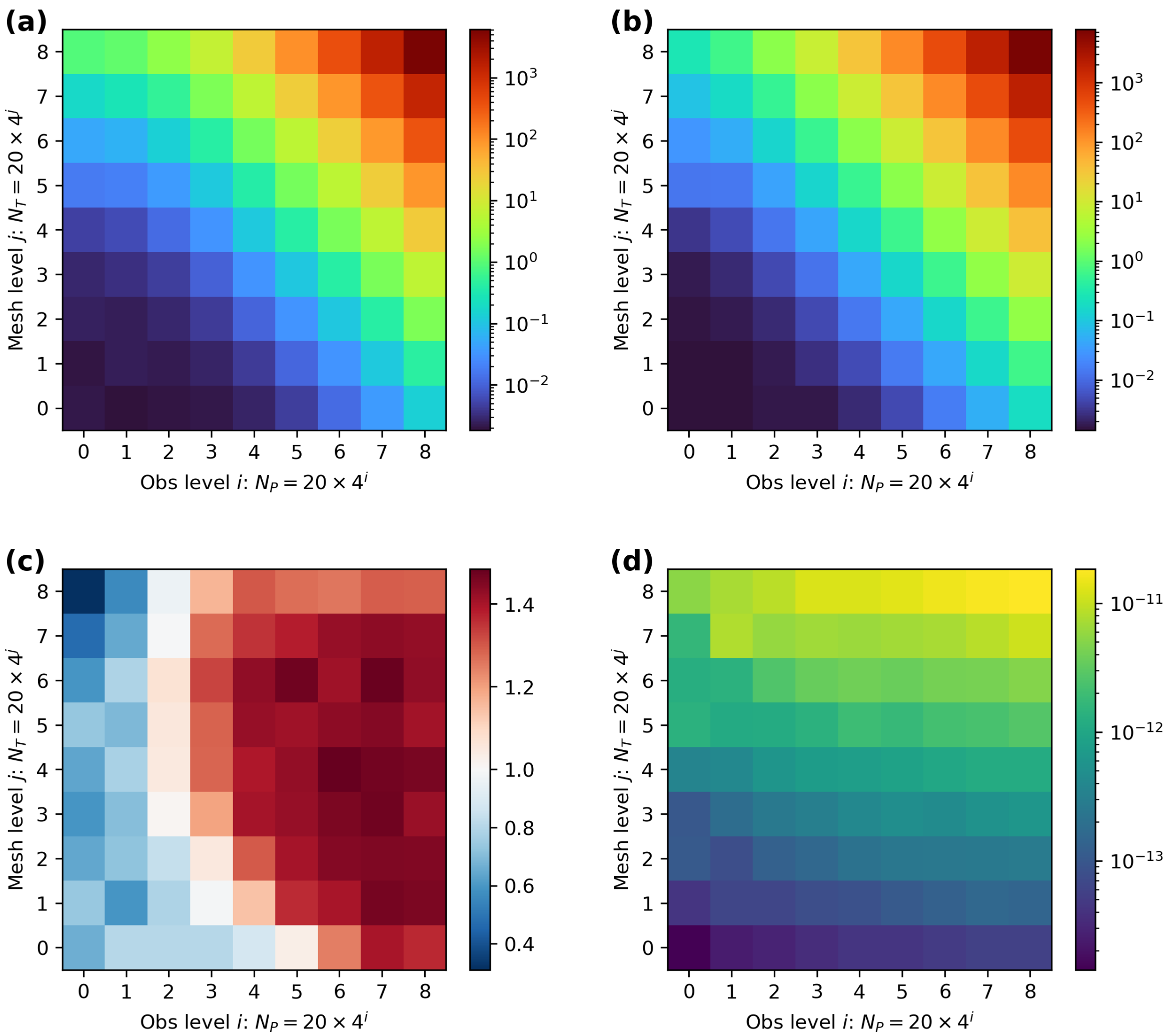

Fig. 1: Benchmark of `WerSch_numba_v1` and `WerSch_numba_v2`. The source body is a unit sphere ($R = 1.0$ km) with constant density $\rho = 2670\ \mathrm{kg/m^3}$, discretized by recursive icosahedral subdivision, yielding $N_T = 20 \times 4^j$ triangular faces for refinement level $j = 0, \ldots, 8$. Observation points are placed on a concentric sphere of radius $r_{\mathrm{obs}} = 1.1$ km using Fibonacci sphere sampling, with $N_P = 20 \times 4^i$ points for level $i = 0, \ldots, 8$. Shown from top-left to bottom-right: (a) runtime of Version 1 (sec), (b) runtime of Version 2, (c) runtime ratio t_2/t_1, and (d) maximum absolute difference in the computed vertical gravity component g_z (mGal) between the two versions.

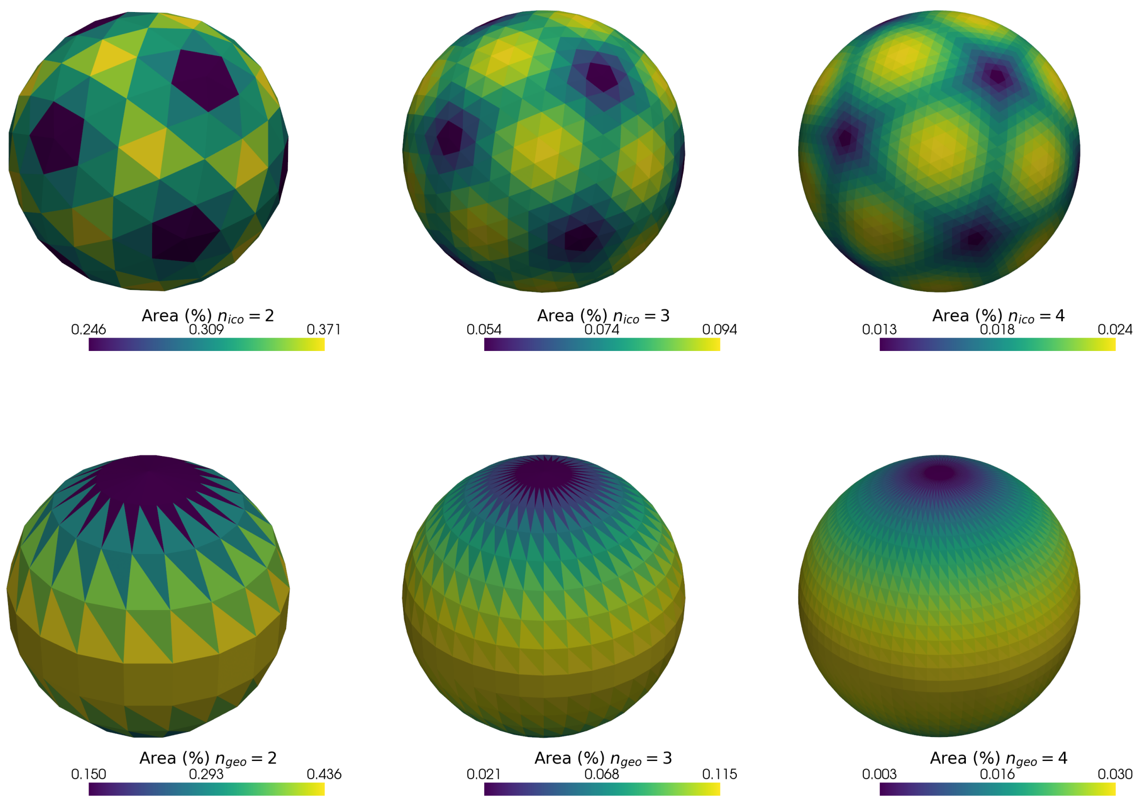

Fig. 2: Two common strategies for polyhedral approximation of a unit sphere: (top row) recursive subdivision of an icosahedron and (bottom row) triangulated regular geographic grid. Meshes at refinement levels $n = 2, 3, 4$ are shown, colored by the relative area of each triangular face (% of total surface area). The icosahedral meshes exhibit more uniform element sizes, while the geographic grids show pronounced polar clustering, leading to high variability in triangle areas-particularly near the poles.

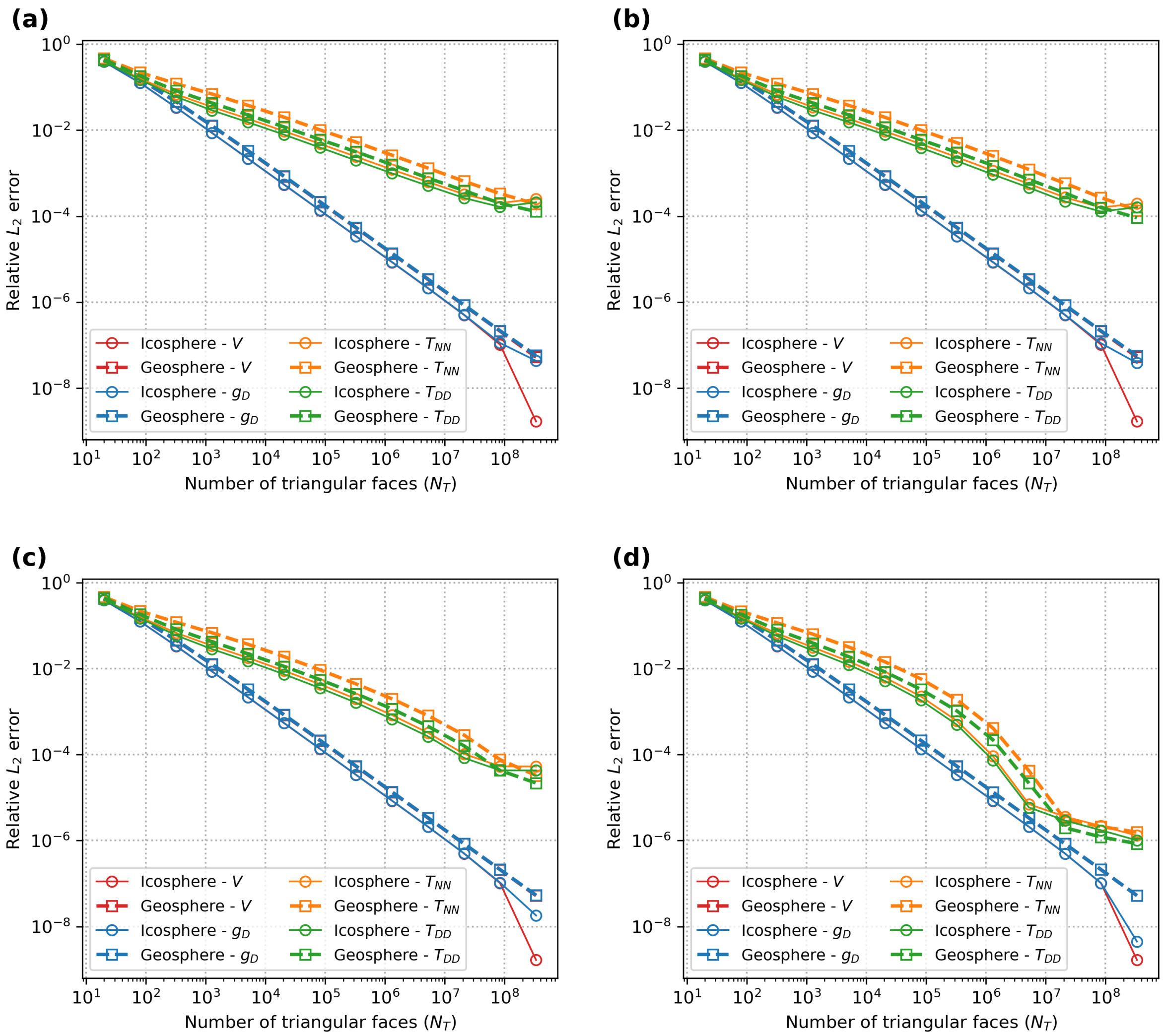

Fig. 3: Convergence of polyhedral gravity solutions toward the analytical field of a homogeneous Earth-sized sphere. Relative L_2 errors in selected field components (potential V, down-component g_D, and diagonal gradients T_{NN}, T_{DD}) are shown as functions of mesh resolution (N_T = number of triangular faces). Panels correspond to different observation altitudes: (a) 0.001 km, (b) 0.1 km, (c) 1 km, and (d) 10 km. Solid lines with open circles represent icosahedral discretizations; dashed lines with open squares denote geographic (longitude–latitude) grids. Both methods converge with increasing resolution, with errors decreasing more rapidly at higher altitudes. The agreement between the two formulations confirms that discrepancies are due to geometric approximation rather than numerical computation.

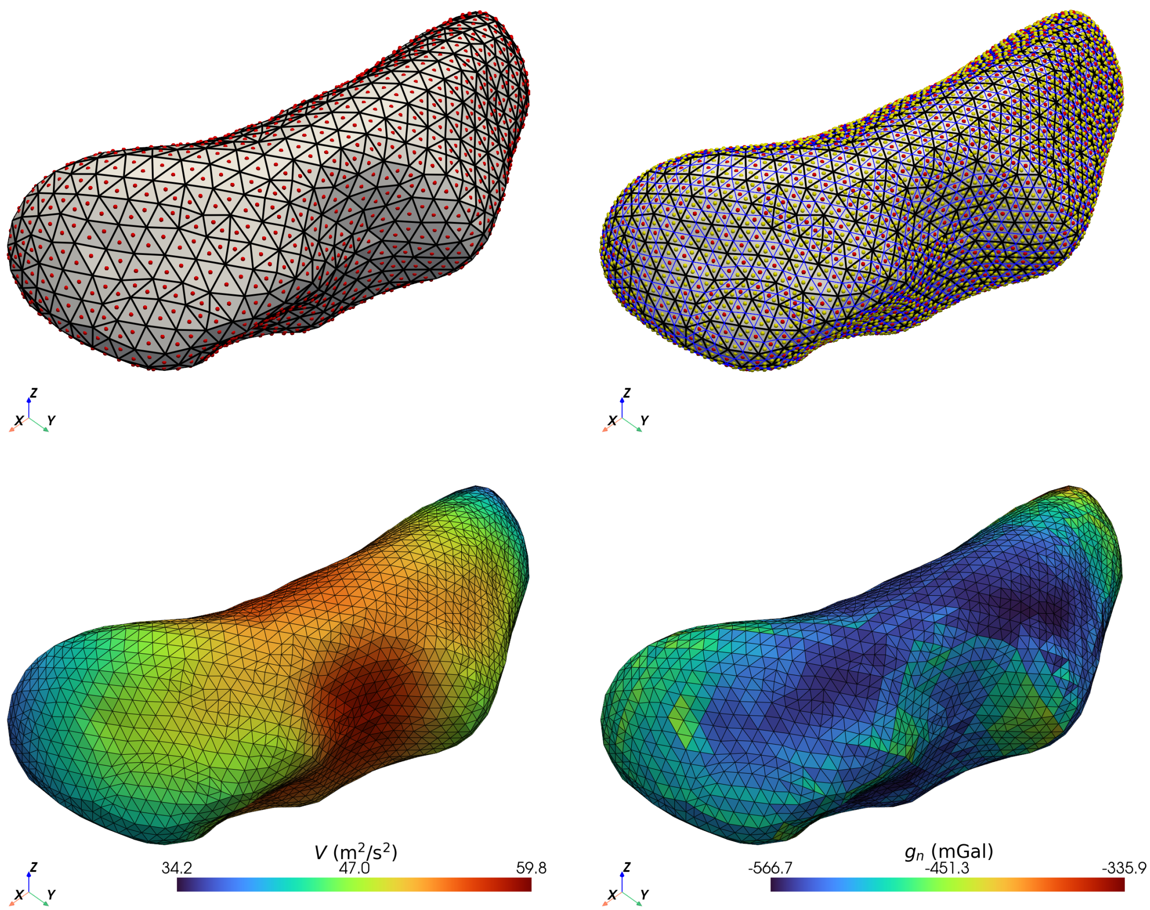

Fig. 4: Linear refinement of the EROS shape model and surface gravity fields. Top-left: original mesh with face centers (red). Top-right: refined mesh showing new interior edges (blue) and new face centers (yellow). Bottom-left: gravitational potential V on refined surface. Bottom-right: normal gravity component g_n.

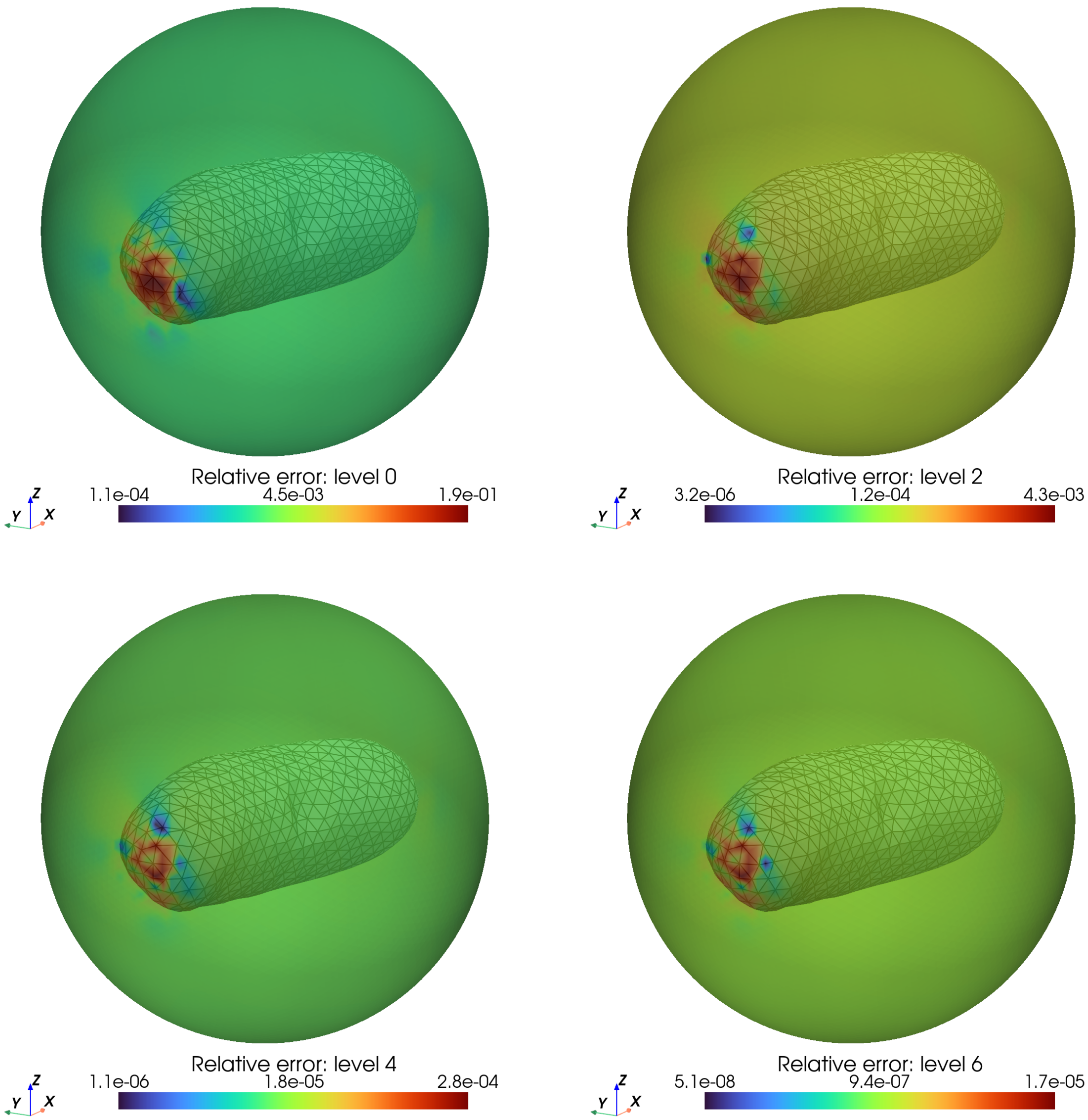

Fig. 5: Pointwise relative errors using Green's third identity approximated gravitational potential on a sphere of radius 17.8 km, shown for four mesh refinement levels. Errors are initially concentrated near the elongated tips of EROS (along $\pm x$) where the surface lies closest to the evaluation sphere, and this spatial pattern persists even at the finest resolution, albeit with dramatically reduced magnitude.

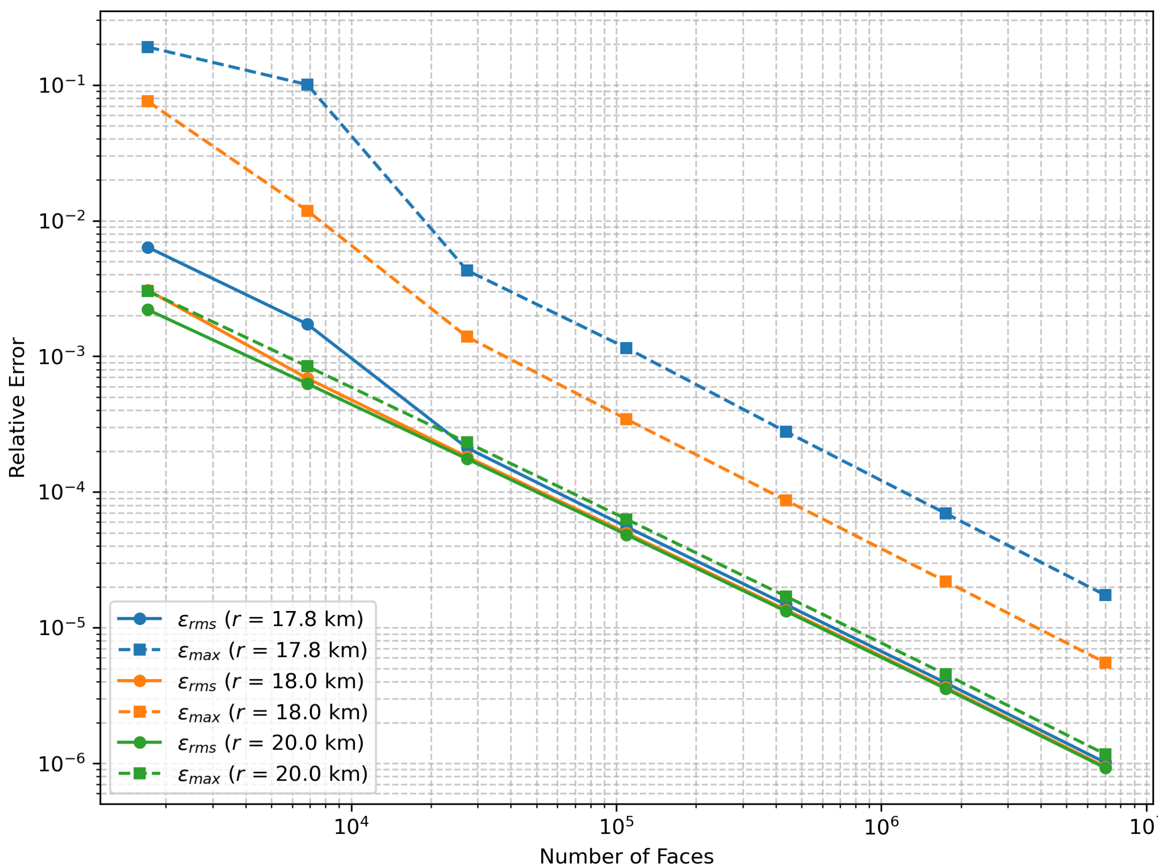

Fig. 6: Convergence of Green's third identity approximation for EROS's exterior gravitational potential. Root Mean Square (RMS) relative error (ε_{rms}, solid lines) and maximum relative error (ε_{max}, dashed lines) in gravitational potential decrease with mesh refinement across three evaluation radii. Errors diminish by over three orders of magnitude as the number of faces increases from 1.7×10^3 (level 0) to 7.0×10^6 (level 6).

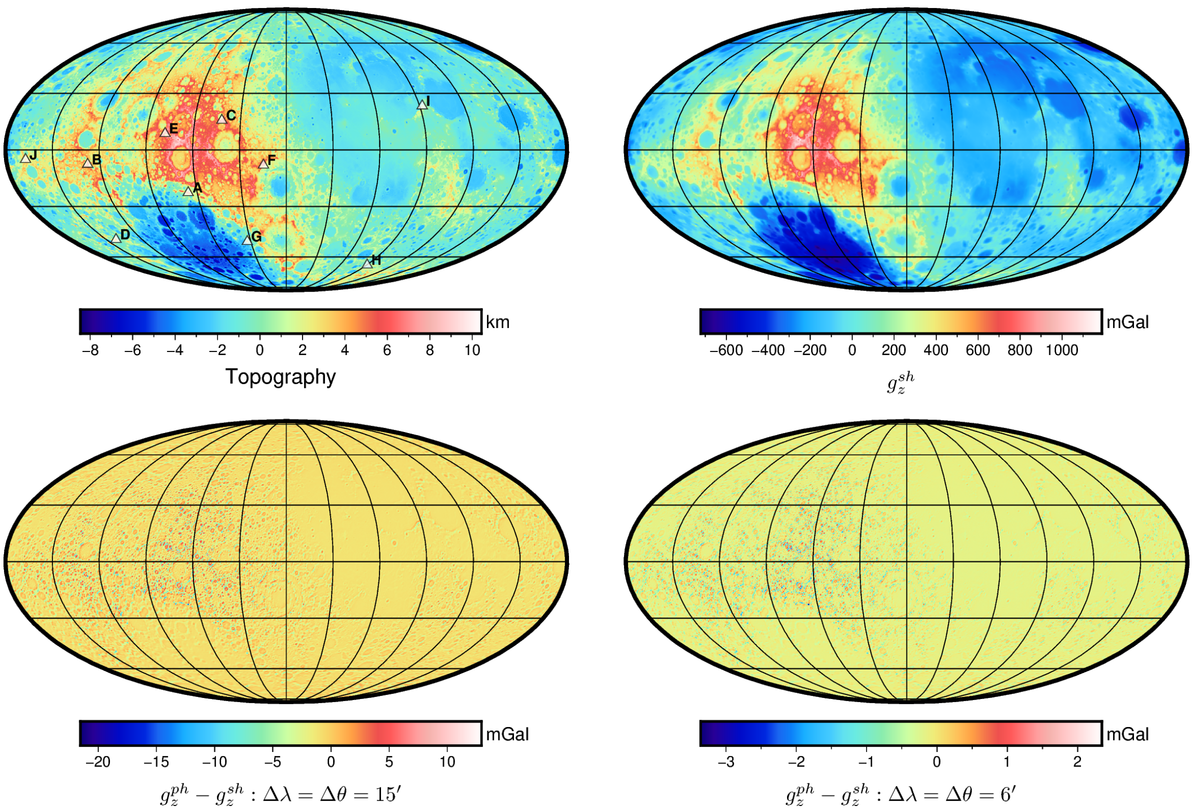

Fig. 7: Global comparison of lunar gravity derived from spherical harmonics (SH) and polyhedral (PH) methods. Top left: Lunar topography with white triangles marking the locations of the largest g_z errors in the 15-arcmin PH-SH difference; triangle sites are separated by at least 1000 km. Top right: Downward gravity component g_z^{sh} from the SH model. Bottom left: Difference $g_z^{\mathrm{ph}} - g_z^{\mathrm{sh}}$ for the 15-arcmin PH model. Bottom right: Same difference for the 6-arcmin PH model.

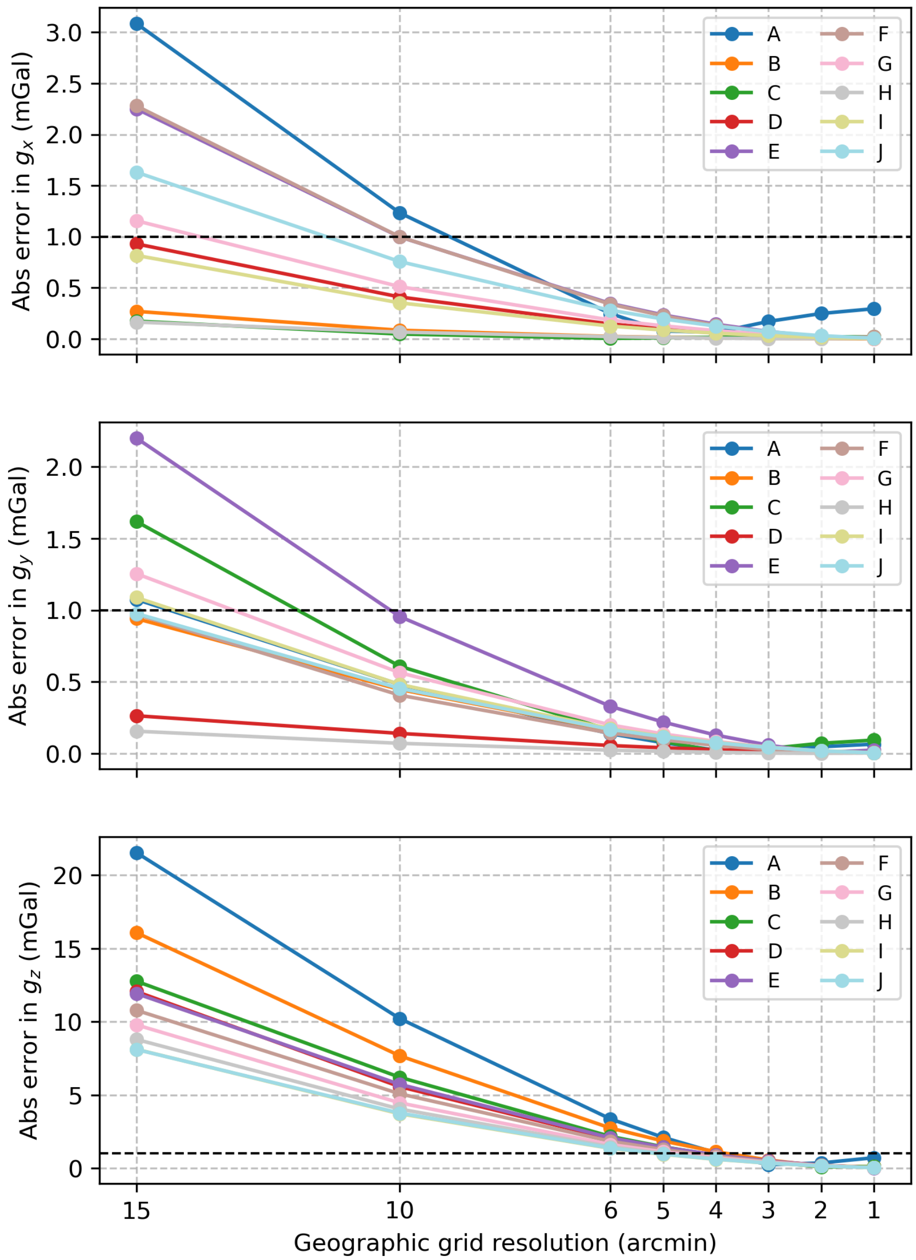

Fig. 8: Absolute errors in the gravity components (g_x, g_y, g_z) at the ten labelled sites (A-J) identified in Figure 7, plotted against polyhedral mesh resolution from 15 to 1 arcminute. Errors are relative to the spherical harmonic (SH) reference solution. The dashed horizontal line indicates the 1 mGal level.

Table 1: Comparison of recursive icosahedral subdivision and triangulated regular geographic grids for approximating an ideal unit sphere. For each refinement level n (with corresponding numbers of vertices N_V and triangular faces N_T), the table reports relative surface area and volume discretization errors (ϵ_{area}, ϵ_{vol}) and the shortest and longest edge lengths (ψ_{min}, ψ_{max}), expressed as spherical distances in arcminutes, across levels $n = 0$ to 12.

n	N_V	N_T	**Icosahedron**				**Geographic**			
			$\epsilon^{\mathrm{ico}}_{\mathrm{area}}$	$\epsilon^{\mathrm{ico}}_{\mathrm{vol}}$	$\psi^{\mathrm{ico}}_{\mathrm{min}}$	$\psi^{\mathrm{ico}}_{\mathrm{max}}$	$\epsilon^{\mathrm{geo}}_{\mathrm{area}}$	$\epsilon^{\mathrm{geo}}_{\mathrm{vol}}$	$\psi^{\mathrm{geo}}_{\mathrm{min}}$	$\psi^{\mathrm{geo}}_{\mathrm{max}}$
0	12	20	2.4e-01	4.0e-01	3.8e+03	3.8e+03	2.5e-01	4.3e-01	3.6e+03	5.5e+03
1	42	80	7.2e-02	1.3e-01	1.9e+03	2.2e+03	8.0e-02	1.5e-01	1.3e+03	3.0e+03
2	162	320	1.9e-02	3.4e-02	8.7e+02	1.1e+03	2.3e-02	4.6e-02	3.7e+02	1.6e+03
3	642	1280	4.8e-03	8.6e-03	4.1e+02	5.7e+02	6.3e-03	1.3e-02	9.9e+01	8.3e+02
4	2562	5120	1.2e-03	2.2e-03	2.0e+02	2.8e+02	1.7e-03	3.3e-03	2.6e+01	4.2e+02
5	10242	20480	3.0e-04	5.4e-04	9.8e+01	1.4e+02	4.2e-04	8.4e-04	6.5e+00	2.1e+02
6	40962	81920	7.5e-05	1.4e-04	4.8e+01	7.1e+01	1.1e-04	2.1e-04	1.6e+00	1.1e+02
7	163842	327680	1.9e-05	3.4e-05	2.4e+01	3.6e+01	2.7e-05	5.3e-05	4.1e-01	5.4e+01
8	655362	1310720	4.7e-06	8.5e-06	1.2e+01	1.8e+01	6.7e-06	1.3e-05	1.0e-01	2.7e+01
9	2621442	5242880	1.2e-06	2.1e-06	6.0e+00	8.9e+00	1.7e-06	3.4e-06	2.6e-02	1.4e+01
10	10485762	20971520	2.9e-07	5.3e-07	3.0e+00	4.4e+00	4.2e-07	8.4e-07	6.5e-03	6.8e+00
11	41943042	83886080	6.3e-08	1.3e-07	1.5e+00	2.2e+00	1.0e-07	2.1e-07	1.6e-03	3.4e+00
12	167772162	335544320	2.5e-08	3.3e-08	7.5e-01	1.1e+00	2.2e-08	5.2e-08	4.0e-04	1.7e+00

Table 2: Statistical comparison of lunar topographic gravity vector components (in mGal) between spherical harmonic (SH) and polyhedral (PH) methods at $15'$ and $6'$ mesh resolutions. The components g_x, g_y, and g_z represent the northward, eastward, and downward directions, respectively, in a local north-oriented coordinate system. Differences are defined as PH − SH.

Method	Component	min	max	mean	std
SH	g_x	−488.8082	933.4147	10.3820	154.3514
	g_y	−551.9941	606.5575	0.0197	150.2317
	g_z	−721.8590	1189.4579	−35.0952	246.4136
PH ($15'$)	g_x	−486.9639	926.2217	10.3856	154.2915
	g_y	−548.6204	605.3110	0.0197	150.2471
	g_z	−721.7818	1182.6341	−36.0165	246.2114
Diff ($15'$)	g_x	−12.0679	12.5141	0.0033	0.8889
	g_y	−11.6858	10.1128	−0.0000	0.7200
	g_z	−21.5349	12.9305	−0.9188	1.1335
PH ($6'$)	g_x	−488.5107	932.2260	10.3829	154.4438
	g_y	−551.4422	606.3592	0.0197	150.3391
	g_z	−721.8638	1187.8352	−35.2429	246.3798
Diff ($6'$)	g_x	−1.9898	2.0846	0.0006	0.1456
	g_y	−2.1117	1.6070	−0.0000	0.1181
	g_z	−3.3589	2.3430	−0.1474	0.1888

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